OXYTOCIN

High-alert drug

Hyperstimulation & water intoxication

CHEMICAL NAME Cyclic nonapeptide: Cys-Tyr-Ile-Gln-Asn-Cys-Pro-Leu-Gly-NH_2 (disulfide bridge Cys1–Cys6) — synthetic

CLASS Uterotonic hormone (posterior pituitary)

TRADE NAMES Syntocinon · Pitocin · Syntometrine (with ergometrine)

THERAPEUTIC USES / INDICATIONS

- Induction and augmentation of labour.
- Active management of the third stage (prevent PPH).
- Treatment of postpartum haemorrhage (uterine atony).
- Incomplete/missed miscarriage (with other agents); caesarean section (after delivery of baby).

PHARMACODYNAMICS

- Stimulates **oxytocin receptors** in myometrium → ↑ frequency and force of uterine contractions (sensitivity rises near term).
- Milk ejection (let-down); weak antidiuretic (ADH-like) effect at high doses.

ADMINISTRATION

- **PPH prevention:** IM 10 units after delivery (US label: after the placenta).
- **Induction/augmentation:** dilute IV infusion via syringe/volumetric pump (UK: 5 units in 500 mL saline), start low and increase at intervals of ≥20 min.
- **Caesarean:** IV infusion over 5 min, not a rapid bolus (hypotension, tachycardia).
- Store in fridge (2–8 °C).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting.
- Headache, flushing.
- Hypotension and reflex tachycardia (rapid IV bolus).

ADVERSE EFFECTS (SERIOUS)

- Uterine hyperstimulation (>5 contractions in 10 min) → fetal distress, uterine rupture.
- Water intoxication / hyponatraemia (prolonged high-dose infusion in hypotonic fluids) → seizures.
- Amniotic fluid embolism (rare); postpartum haemorrhage after prolonged use (atony).

DOSAGES

- **PPH prevention:** 10 units IM (US label) or 5 units IV infusion over 5 min after the placenta (UK label).
- **PPH treatment:** 5 units IV over 5 min, then in severe cases 5–20 units in 500 mL infusion run to control uterine atony.
- **Induction:** start 1–4 milliunits/min, increase by ≤1–2 mU/min at intervals of ≥20 min; max 20 mU/min (UK label).
- **Caesarean:** 5 units IV infusion over 5 min after delivery.

CONTRAINDICATIONS & CAUTIONS

- Hypertonic uterine contractions, mechanical obstruction to delivery, fetal distress when delivery not imminent.
- Cephalopelvic disproportion, malpresentation, placenta praevia, vasa praevia, cord presentation.
- Previous classical caesarean scar (relative).

KEY INTERACTIONS

- **Prostaglandins:** potentiate effect; wait ≥6 h after last prostaglandin before oxytocin.
- **Vasoconstrictors, anaesthetics:** hypertension or hypotension.

PHARMACOKINETICS

- **Onset:** IV immediate; IM 3–5 min. **Duration:** IV ~20 min after stopping; IM 2–3 h.
- Metabolised by oxytocinases in liver, kidney, placenta.
- **t½:** 3–5 min.

NURSING MANAGEMENT

- Continuous CTG and contraction monitoring (frequency, duration, strength) during infusion.
- Reduce/stop infusion for hyperstimulation or abnormal fetal heart rate; call obstetrician.
- Monitor maternal BP, HR, fluid balance; use isotonic fluids.
- Use separate dedicated line with pump; document rate changes.
- Postpartum: monitor uterine tone, lochia, BP.

PATIENT EDUCATION

- Explain that contractions will become stronger and more regular.
- Report severe constant pain or feeling unwell.
- Postpartum: report heavy bleeding.

KEY POINTS

- Uterotonic — induction, third stage, PPH.
- Infusion by pump with continuous CTG.
- Third stage 10 units IM.
- Hyperstimulation, rupture, water intoxication.

SOURCES CHECKED **[665]** Oxytocin 10 IU/ml Concentrate for Solution for Infusion (UK SmPC) **[666]** Oxytocin, IM, IV (US label) **[667]** Oxytocin, IM, IV (US label, Fresenius Kabi USA)

Full references p. 432–457